

Question 2

Q: Given a real-world scene captured under low light, describe how the image sensing and acquisition process influences the final digital image.

Theoretical Concept: Low Light Acquisition

In low-light conditions, the number of photons reaching the camera sensor (CCD/CMOS) is reduced. This leads to a low **Signal-to-Noise Ratio (SNR)**. To compensate, cameras increase ISO sensitivity, which amplifies both the light signal and the inherent electrical noise, resulting in a grainy digital image.

Correction: We can improve low-light images using Histogram Equalization or Gamma Correction to enhance contrast, followed by a smoothing filter to reduce noise.

OpenCV Python Solution

 Copy

```
import cv2 # Import OpenCV for image processing algorithms
import numpy as np # Import Numpy for mathematical matrix operations
import matplotlib.pyplot as plt # Import Matplotlib to visualize the output on screen

# Step 1: Create a synthetic image representing a clear scene
img = np.zeros((100, 100), dtype=np.uint8) # Create a blank black image array filled with zeros
cv2.circle(img, (50, 50), 30, 200, -1) # Draw a filled or outlined circle on the image

# Step 2: Simulate low light by drastically reducing contrast and brightness
low_light = cv2.convertScaleAbs(img, alpha=0.2, beta=0) # Scale and shift pixel values, clamp to 0-255
```

```
# Step 3: Apply Histogram Equalization to mathematically stretch the contrast
enhanced = cv2.equalizeHist(low_light) # Enhance contrast using histogram equalization

# Step 4: Display the final enhanced image
plt.figure(figsize=(8,4)) # Create a new figure window for display
plt.subplot(121), plt.imshow(low_light, cmap='gray'), plt.title('Low Light Image') # Place this image in a subplot
plt.subplot(122), plt.imshow(enhanced, cmap='gray'), plt.title('Enhanced via Equalization') # Place this image in
plt.show() # Show the final plot window to the user
```

Expected Output

The output will show a side-by-side plot. The left image will be very dark and hard to see, while the right image will have its brightness perfectly restored by the OpenCV equalization algorithm.

How to Execute & Submit

1. Google Colab Execution:

- Open [Google Colab](#) and create a New Notebook.
- Click the  **Copy** button on the code block above.
- Paste it into a Colab cell and press **Shift + Enter** to run.

2. Uploading to GitHub:

- In Colab, go to **File > Save a copy in GitHub**.
- Authorize your GitHub account.

- Select your Computer Vision Lab repository and click **OK** to commit the code.